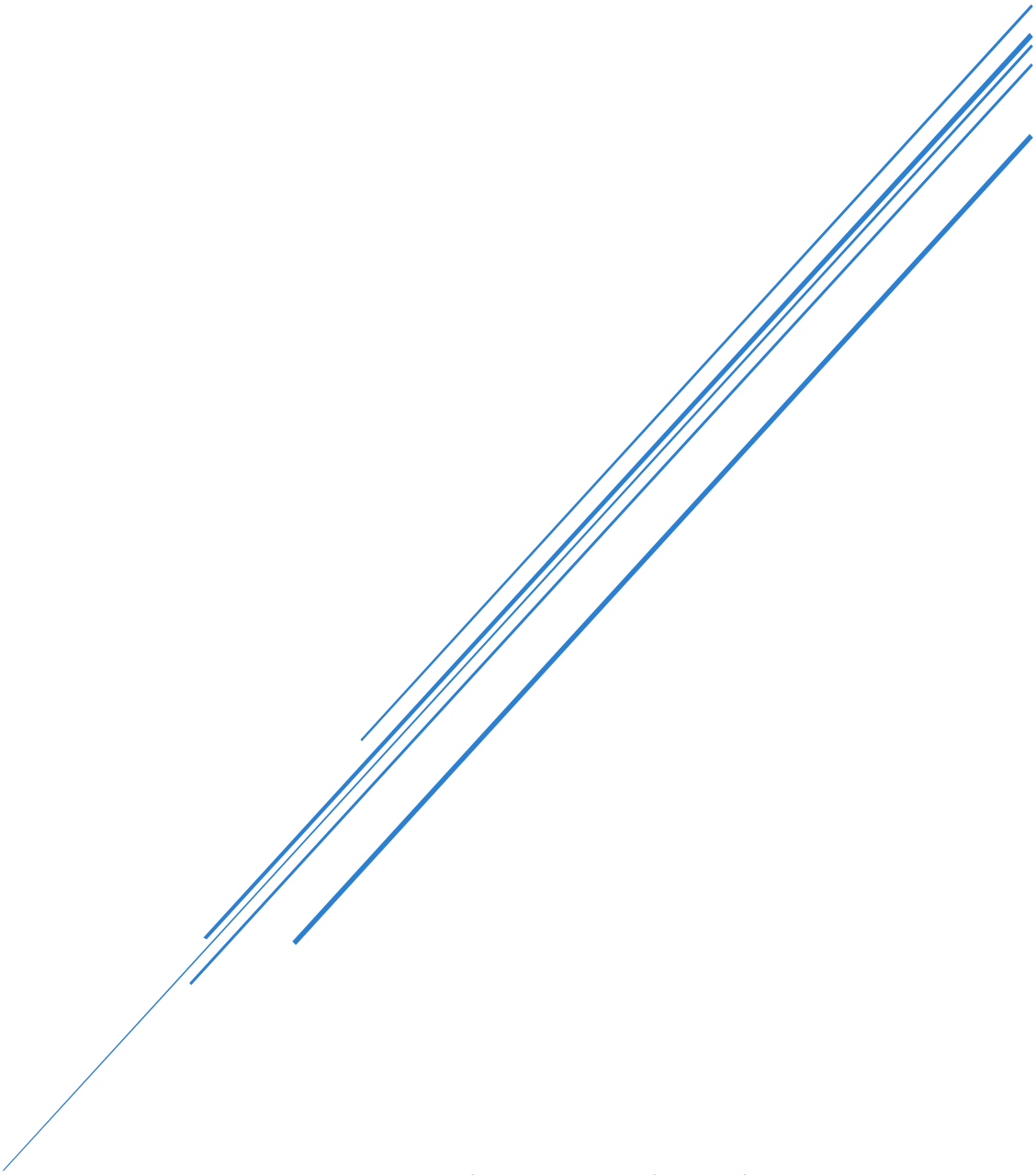
LAB 8 ANSWERS REPORT

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Flinders University – Tonsley

Artificial Intelligence (COMP3742)

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Question 1

Define a Prolog knowledge base for the Genus Panthera. Refer to the Panthera cladogram on the previous page and the Wikipedia site https://en.wikipedia.org/wiki/Panthera

Answer found in Source/panthera.pl

Question 2

Define Prolog rules and facts to describe the Panthera cladogram in the form of a Prolog Knowledge base

Answer found in Source/panthera.pl

Question 3

Query the node two generations prior (grandparent node) of “Cave Lion”. Ensure to show the query and results

Query:

```
parent(Grandparent, P), parent(P, cave_lion).
```

Result:

```
Grandparent = ancestor4,
P = ancestor5.
```

Question 4

Find the direct descendants (child nodes) of “Ancestor 4”. Ensure to show the query and results

Query:

```
parent(ancestor4, C).
```

Result:

```
C = lion;
C = ancestor5.
```

Question 5

Query the immediate ancestral node (parent node) of ‘Lion”. Ensure to show the query and results

Query:

```
parent(P, lion).
```

Result:

```
P = ancestor4.
```